

Supervised Machine Learning: Regression - Final Assignment

Import the required libraries

```
In [30]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, PolynomialFeatures, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score
```

Importing the Dataset

I choose Medical cost Personal Datasets (<https://www.kaggle.com/datasets/mirchoi0218/insurance/data>). It allows us to explore the factors influencing individual medical costs. This dataset contains relevant information such as age, gender, body mass index (BMI), number of children, smoking status and geographic region, as well as associated medical costs. It is particularly interesting for linear regression analysis, because it allows us to study how these variables influence health care costs. This analysis could provide valuable insights for insurance companies, hospitals and policymakers to optimize medical risk and cost management.

```
In [31]: data = pd.read_csv('insurance.csv')
data.info()
data.head()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column  Non-Null Count  Dtype  
---  --
 0   age      1338 non-null     int64  
 1   sex      1338 non-null     object  
 2   bmi      1338 non-null     float64 
 3   children 1338 non-null     int64  
 4   smoker   1338 non-null     object  
 5   region   1338 non-null     object  
 6   charges  1338 non-null     float64 
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB

Out[31]:
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

1. About the Data

We will start to clean data

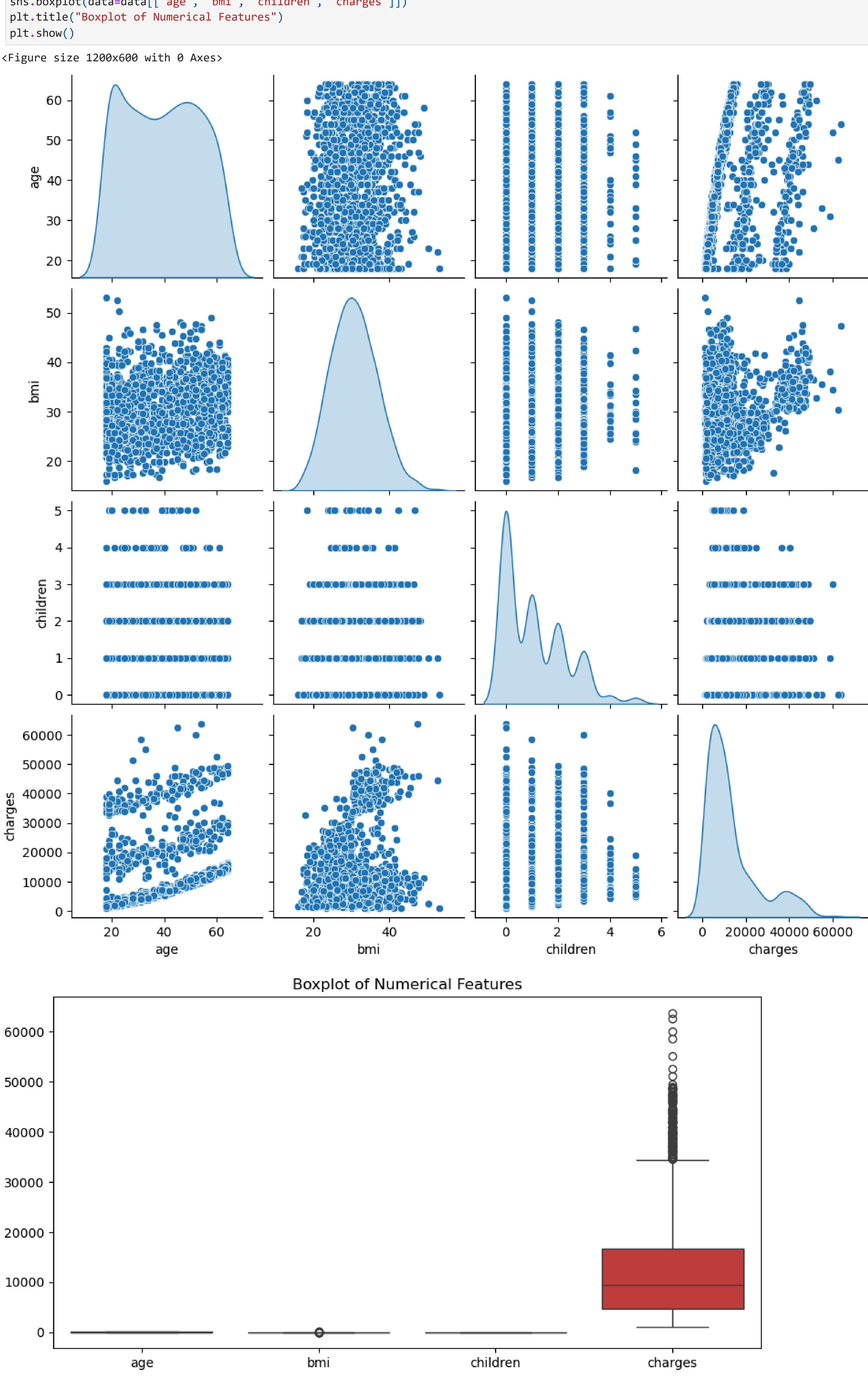
```
In [32]: # Checking for missing values
print("Missing values:")
print(data.isnull().sum())

Missing values:
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64

No missing value, let's check visually the consistency.
```

```
In [33]: # Visualizing distributions of numerical variables
plt.figure(figsize=(12, 6))
sns.pairplot(data, diag_kind='kde')
plt.show()

# Checking for outliers using boxplots
plt.figure(figsize=(10, 5))
sns.boxplot(data=data[['age', 'bmi', 'children', 'charges']])
plt.title("Boxplot of Numerical Features")
plt.show()
```



There doesn't seem to be any outliers. Let's encode and scale the other data for easier processing.

```
In [34]: # Encoding categorical variables
categorical_features = ['sex', 'smoker', 'region']
numeric_features = ['age', 'bmi', 'children']

ohe = OneHotEncoder(drop='first')
scaler = StandardScaler()

In [35]: preprocessor = ColumnTransformer([
    ('num', scaler, numeric_features),
    ('cat', ohe, categorical_features)
])

# Apply preprocessing
X = data.drop(columns=['charges'])
y = data['charges']
X = pd.DataFrame(preprocessor.fit_transform(X))

# Splitting dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Our data are ready to use for linear regression. We have 20% for test and 80% for train. Data are randomised to avoid bias.

2. Objectives

The main objective of this analysis is to **explore the factors influencing medical insurance costs** and develop predictive models that can accurately estimate charges based on individual attributes.

This study aims to determine whether a linear relationship exists between medical costs and other factors such as age, BMI, smoking status, and region, or if more complex interactions must be captured through polynomial and regularized regression.

By comparing different regression techniques, **we seek to identify the most effective model for both interpretability** (understanding key cost drivers) **and prediction** (minimizing error in cost estimates), which can be valuable for insurance companies and healthcare policymakers.

3. Linear Regression Models

```
In [36]: # Simple Linear Regression (BaseLine)

simple_model = LinearRegression()
simple_model.fit(X_train, y_train)
y_pred_simple = simple_model.predict(X_test)

# Evaluation
simple_mse = mean_squared_error(y_test, y_pred_simple)
simple_r2 = r2_score(y_test, y_pred_simple)

In [37]: # Polynomial Regression

poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
y_pred_poly = poly_model.predict(X_test_poly)

# Evaluation
poly_mse = mean_squared_error(y_test, y_pred_poly)
poly_r2 = r2_score(y_test, y_pred_poly)

In [38]: # Regularized Regression (Ridge & Lasso)

ridge_model = Ridge(alpha=1.0)
ridge_model.fit(X_train, y_train)
y_pred_ridge = ridge_model.predict(X_test)

ridge_mse = mean_squared_error(y_test, y_pred_ridge)
ridge_r2 = r2_score(y_test, y_pred_ridge)

lasso_model = Lasso(alpha=0.1)
lasso_model.fit(X_train, y_train)
y_pred_lasso = lasso_model.predict(X_test)

lasso_mse = mean_squared_error(y_test, y_pred_lasso)
lasso_r2 = r2_score(y_test, y_pred_lasso)

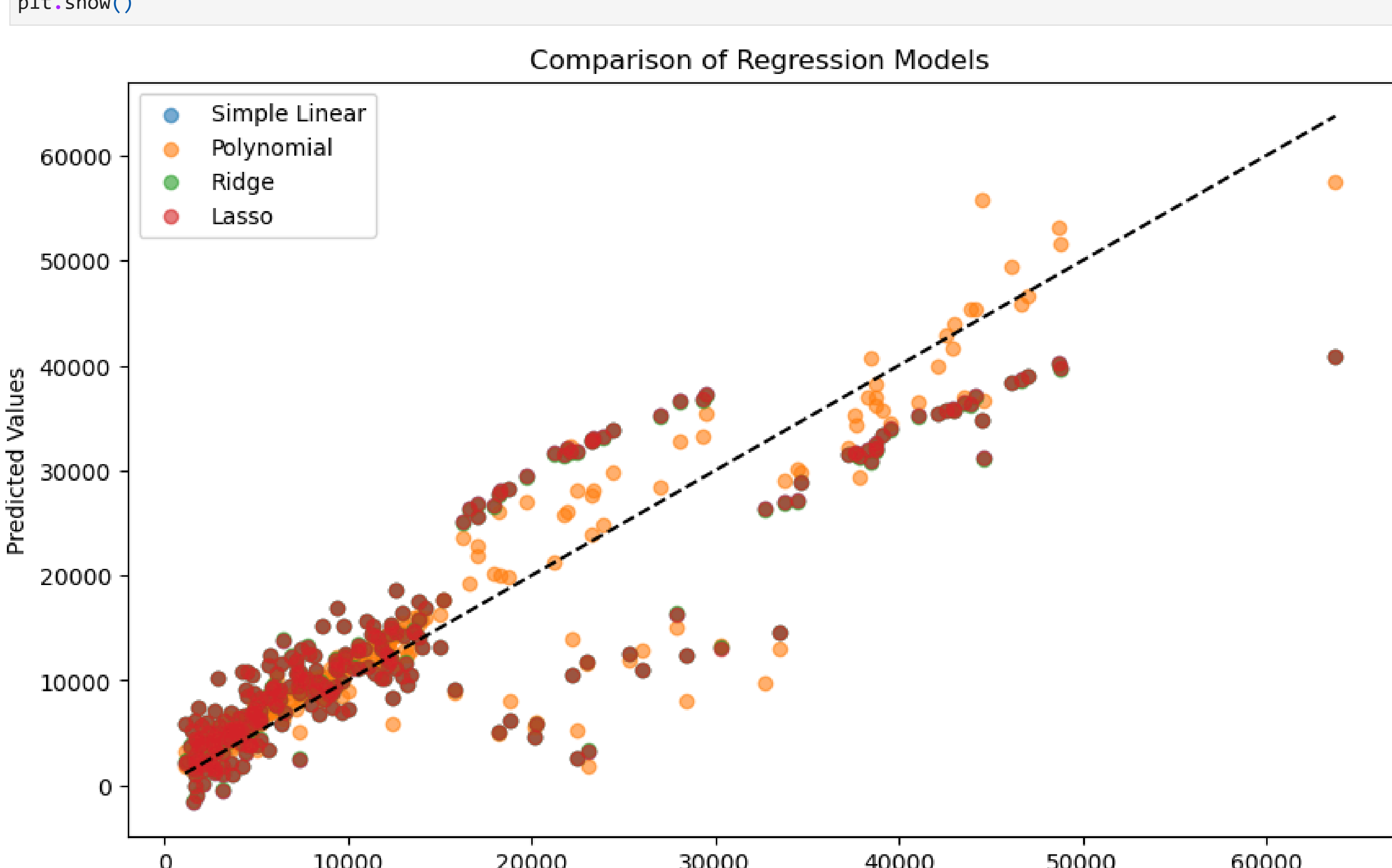
In [39]: # Comparing Model Performance

models_results = pd.DataFrame({
    "Model": ["Simple Linear Regression", "Polynomial Regression", "Ridge Regression", "Lasso Regression"],
    "MSE": [simple_mse, poly_mse, ridge_mse, lasso_mse],
    "R² Score": [simple_r2, poly_r2, ridge_r2, lasso_r2]
})

print(models_results)
```

	Model	MSE	R² Score
0	Simple Linear Regression	3.359692e+07	0.783593
1	Polynomial Regression	2.071281e+07	0.866583
2	Ridge Regression	3.364539e+07	0.783281
3	Lasso Regression	3.359779e+07	0.783587

```
In [40]: # Visualizing predictions
plt.figure(figsize=(10, 6))
plt.scatter(y_test, y_pred_simple, label="Simple Linear", alpha=0.6)
plt.scatter(y_test, y_pred_poly, label="Polynomial", alpha=0.6)
plt.scatter(y_test, y_pred_ridge, label="Ridge", alpha=0.6)
plt.scatter(y_test, y_pred_lasso, label="Lasso", alpha=0.6)
plt.plot([min(y_test), max(y_test)], [min(y_test), max(y_test)], color="black", linestyle="dashed")
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.legend()
plt.title("Comparison of Regression Models")
plt.show()
```



4. Insights and key findings

The analysis of different regression models on the Medical Cost Personal Datasets reveals interesting insights into their predictive capabilities:

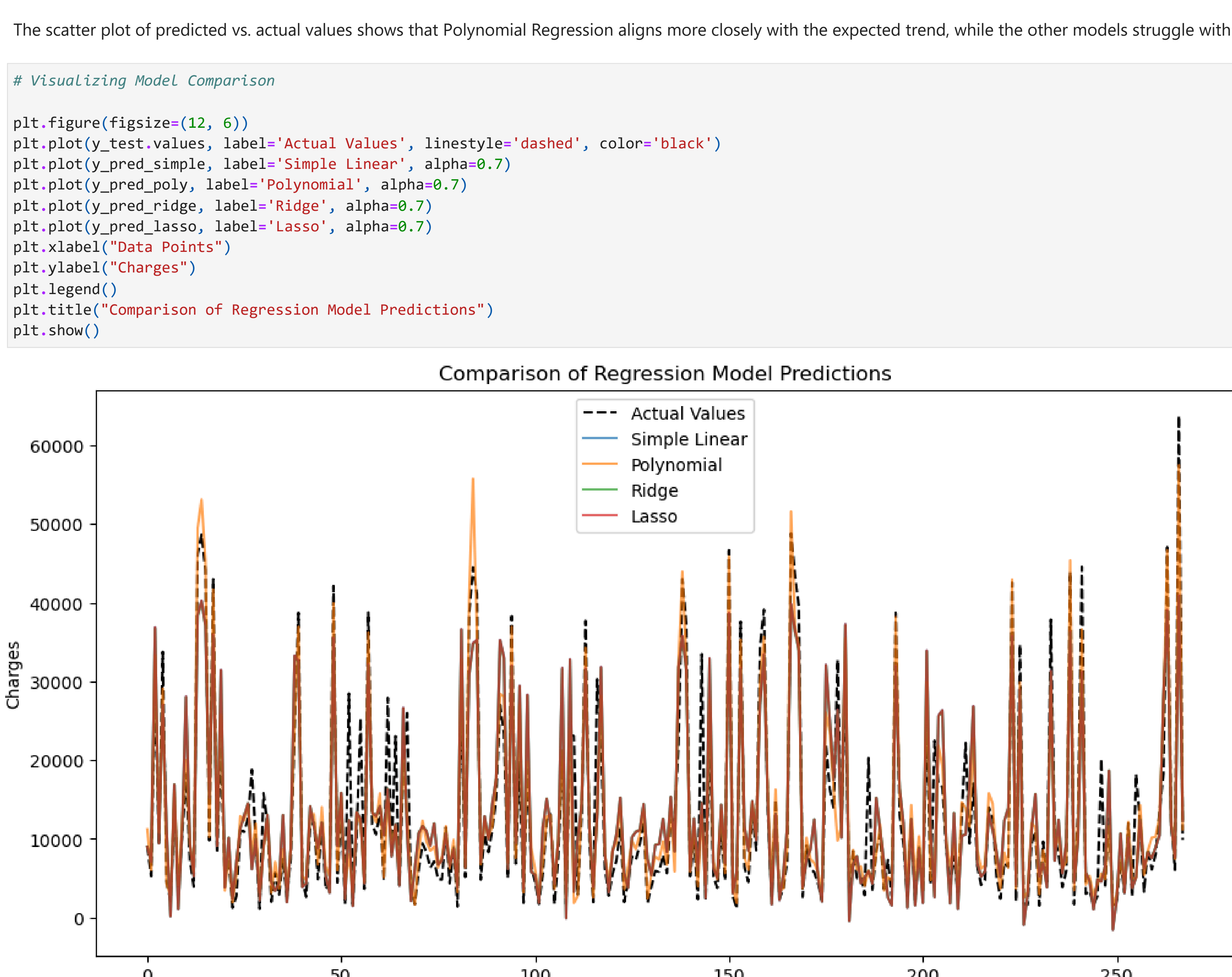
Polynomial Regression significantly outperforms the other models with an **MSE of ~20.7M** and an **R² score of 0.867**, suggesting that introducing non-linearity better captures the complex relationships within the data.

Simple Linear Regression, Ridge, and Lasso perform similarly, with an **MSE of ~33.6M** and an **R² score of ~0.784**, indicating that a basic linear approach might not fully capture the variations in medical charges.

Ridge and Lasso Regression do not provide significant improvements over the simple linear model, implying that feature selection and regularization do not heavily impact this dataset.

The scatter plot of predicted vs. actual values shows that Polynomial Regression aligns more closely with the expected trend, while the other models struggle with higher charge values

```
In [41]: # Visualizing Model Comparison
```



5. Next Steps

To further improve the predictive power and reliability of our model, several enhancements can be considered.

First, **feature engineering** can be expanded by exploring interactions between variables, such as BMI and smoking status, which may better capture medical cost variations.

Additionally, hyperparameter tuning for Ridge and Lasso regressions could **improve regularization effects, reducing overfitting**.

Incorporating non-linear models, such as decision trees or gradient boosting, may provide a better fit for complex patterns.

Finally, **acquiring more diverse data** or incorporating additional healthcare-related features, such as pre-existing conditions or lifestyle factors, could significantly enhance model accuracy and interpretability.